Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: River Rock Ranches (Formerly Yosemite Jersey Dairy)

Physical address of dairy:

Number and Street	City	County	Zip Code
21522 Geer AVE	Hilmar	Merced	95324

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1995

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X200-X069-XXXX

B. OPERATORS

California Dairy Farms-Bloss

Operator name: California Dairy Farms-Bloss

Telephone no.: (209) 285-0001 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
6071 Larson AVE	Hilmar	CA	95324

This operator is responsible for paying permit fees.

C. OWNERS

Fagundes, Fagundes, Fagundes

Legal owner name: Fagundes, Fagundes, Fagundes

Telephone no.: (209) 383-6075 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2717	Merced	CA	95324

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	500
Number under roof	1,840	464	750	0	500	0
Maximum number	1,840	464	750	0	500	500
Average number	1,840	464	750	0	500	500
Avg live weight (lbs)	1,000	1,050	860	0		

Predominant milk cow breed: Jersey

Average milk production: 51 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 60,067.62 *tons per reporting period*

Total nitrogen from manure: 770,409.66 *lbs per reporting period*

After ammonia losses (30% loss applied): 539,286.76 *lbs per reporting period*

Total phosphorus from manure: 121,443.69 *lbs per reporting period*

Total potassium from manure: 329,408.26 *lbs per reporting period*

Total salt from manure: 973,060.80 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 25,556,400 *gallons*

Total nitrogen generated: 79,947.20 *lbs*

Total phosphorus generated: 11,706.40 *lbs*

Total potassium generated: 122,651.54 *lbs*

Total salt generated: 738,032.80 *lbs*

	25,556,400	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	25,556,400	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
BLEVINS	17	17	2	process wastewater	X045-X020-X062-XXXX
BUNGARD	12	12	2	process wastewater	X045-X020-X007-XXXX
ELMQUIST	24	24	2	process wastewater	X045-X020-X004-XXXX
MCCULLOUGH	20	20	2	process wastewater	X045-X020-X076-XXXX
NEWMAN	27	27	2	process wastewater	X045-X020-X079-XXXX
REDDING EAST	9	9	2	process wastewater	X045-X020-X069-XXXX
REDDING WEST	18	18	2	process wastewater	X045-X020-X069-XXXX
TELL	25	25	2	process wastewater	X045-X020-X019-XXXX
WRIGHTS	7	7	2	process wastewater	X045-X020-X079-XXXX
Totals for areas that were used for application	159	159	18		
Totals for areas that were not used for application					
Land application area totals	159	159	18		

B. CROPS AND HARVESTS

BLEVINS

Field name: BLEVINS

11/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 11/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	255.00 *ton*	As-is		69.9	5,700.00	1,400.00	16,000.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.00	171.00	42.00	480.00	939.12

BLEVINS

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	442.00 *ton*	As-is		70.0	4,600.00	600.00	5,600.00		7.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	239.20	31.20	291.20	1,232.40

BUNGARD

Field name: BUNGARD

11/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 12 Plant date: 11/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	194.00 *ton*	As-is		70.0	6,500.00	1,200.00	16,500.00		9.47

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.17	210.17	38.80	533.50	918.59

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	285.60 *ton*	As-is		68.6	4,400.00	600.00	6,100.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.80	209.44	28.56	290.36	1,120.98

ELMQUIST

Field name: ELMQUIST

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 24 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	310.00 *ton*	As-is		70.1	6,600.00	1,400.00	15,800.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.92	170.50	36.17	408.17	950.07

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 24 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	770.00 *ton*	As-is		69.2	4,300.00	700.00	5,600.00		7.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.08	275.92	44.92	359.33	1,403.20

MCCULLOUGH

Field name: MCCULLOUGH

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	321.00 *ton*	As-is		70.0	6,500.00	1,600.00	16,200.00		11.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.05	208.65	51.36	520.02	1,136.34

MCCULLOUGH

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	482.00 *ton*	As-is		70.4	4,200.00	600.00	4,800.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.10	202.44	28.92	231.36	970.17

NEWMAN

Field name: NEWMAN

11/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 27 Plant date: 11/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	452.00 *ton*	As-is		70.0	6,400.00	1,400.00	17,400.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.74	214.28	46.87	582.58	1,165.16

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 27 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	678.00 *ton*	As-is		67.6	5,100.00	600.00	6,000.00		8.06

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.11	256.13	30.13	301.33	1,311.52

REDDING EAST

Field name: REDDING EAST

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 9 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	164.00 *ton*	As-is		71.9	5,300.00	1,300.00	15,200.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.22	193.16	47.38	553.96	1,126.50

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 9 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	237.00 *ton*	As-is		67.6	5,100.00	600.00	6,000.00		8.06

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.33	268.60	31.60	316.00	1,375.36

REDDING WEST

Field name: REDDING WEST

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	352.00 *ton*	As-is		71.9	5,300.00	1,300.00	15,200.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.56	207.29	50.84	594.49	1,208.92

REDDING WEST

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	553.00 *ton*	As-is		70.6	4,000.00	700.00	5,100.00		7.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.72	245.78	43.01	313.37	1,318.72

TELL

Field name: TELL

11/14/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 25 Plant date: 11/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	393.00 *ton*	As-is		71.9	5,300.00	1,300.00	15,200.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.72	166.63	40.87	477.89	971.81

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	516.00 *ton*	As-is		67.6	5,100.00	600.00	6,000.00		8.06

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	20.64	210.53	24.77	247.68	1,078.00

WRIGHTS

Field name: WRIGHTS

11/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 7 Plant date: 11/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	110.00 *ton*	As-is		70.0	6,400.00	1,400.00	17,400.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.71	201.14	44.00	546.86	1,093.71

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 7 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	181.00 *ton*	As-is		71.4	4,100.00	600.00	5,700.00		7.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.86	212.03	31.03	294.77	1,079.69

NUTRIENT BUDGET

A. LAND APPLICATIONS

BLEVINS - 11/13/2022: Oats, silage-soft dough

Field name: BLEVINS

Crop: Oats, silage-soft dough

Plant date: 11/13/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.23	7.22	63.68	604.16	237,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	292.23	2,768,926.00 *gal*
Application event totals		54.23	7.22	63.68	896.39	

02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.06	3.40	77.93	488.27	162,000.00 *gal*
Application event totals		44.06	3.40	77.93	488.27	

03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	19.58	1.51	34.64	217.01	72,000.00 *gal*
Application event totals		19.58	1.51	34.64	217.01	

04/05/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.37	2.27	51.95	325.51	108,000.00 *gal*
Application event totals		29.37	2.27	51.95	325.51	

04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.57	5.14	117.76	737.83	244,800.00 *gal*
Canal	Surface water	0.00	0.00	0.00	237.16	2,247,145.00 *gal*
Application event totals		66.57	5.14	117.76	974.99	

BLEVINS - 06/01/2023: Corn, silage

Field name: BLEVINS

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	123.36	9.53	218.21	1,367.16	453,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	271.30	2,570,587.00 *gal*
Application event totals		123.36	9.53	218.21	1,638.46	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.10	3.25	74.47	466.57	154,800.00 *gal*
Canal	Surface water	0.00	0.00	0.00	254.79	2,414,153.00 *gal*
Application event totals		42.10	3.25	74.47	721.36	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.06	7.43	31.14	151.62	237,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	238.19	2,256,886.00 *gal*
Application event totals		27.06	7.43	31.14	389.81	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	222.15	2,104,855.00 *gal*
Application event totals		0.00	0.00	0.00	222.15	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	16.40	4.50	18.87	91.89	144,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	216.76	2,053,855.00 *gal*
Application event totals		16.40	4.50	18.87	308.66	

BLEVINS - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	20.09	5.52	23.12	112.57	176,400.00 *gal*
Canal	Surface water	0.00	0.00	0.00	191.68	1,816,202.00 *gal*
Application event totals		20.09	5.52	23.12	304.25	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.46	11.93	50.01	243.52	381,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	178.07	1,687,213.00 *gal*
Application event totals		43.46	11.93	50.01	421.58	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	11.48	3.15	13.21	64.33	100,800.00 *gal*
Canal	Surface water	0.00	0.00	0.00	158.85	1,505,126.00 *gal*
Application event totals		11.48	3.15	13.21	223.18	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	11.48	3.15	13.21	64.33	100,800.00 *gal*
Canal	Surface water	0.00	0.00	0.00	137.24	1,300,317.00 *gal*
Application event totals		11.48	3.15	13.21	201.56	

BUNGARD - 11/13/2022: Oats, silage-soft dough

Field name: BUNGARD

Crop: Oats, silage-soft dough

Plant date: 11/13/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

BUNGARD - 11/13/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.92	4.65	41.01	389.04	108,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	308.87	2,065,800.00 *gal*
Application event totals		34.92	4.65	41.01	697.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.74	2.14	49.07	307.43	72,000.00 *gal*
Application event totals		27.74	2.14	49.07	307.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.83	3.00	68.70	430.40	100,800.00 *gal*
Application event totals		38.83	3.00	68.70	430.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.61	3.21	73.60	461.14	108,000.00 *gal*
Application event totals		41.61	3.21	73.60	461.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.06	2.79	63.79	399.66	93,600.00 *gal*
Application event totals		36.06	2.79	63.79	399.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.74	2.14	49.07	307.43	72,000.00 *gal*
Application event totals		27.74	2.14	49.07	307.43	

BUNGARD - 11/13/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	8.32	0.64	14.72	92.23	21,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	232.89	1,557,660.00 *gal*
Application event totals		8.32	0.64	14.72	325.12	

BUNGARD - 06/01/2023: Corn, silage

Field name: BUNGARD

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.35	5.36	122.67	768.57	180,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	278.26	1,861,056.00 *gal*
Application event totals		69.35	5.36	122.67	1,046.83	

06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.41	4.82	110.40	691.72	162,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	257.65	1,723,212.00 *gal*
Application event totals		62.41	4.82	110.40	949.36	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	23.23	6.38	26.74	130.18	144,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	246.72	1,650,144.00 *gal*
Application event totals		23.23	6.38	26.74	376.90	

BUNGARD - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	17.42	4.78	20.05	97.64	108,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	227.86	1,523,976.00 *gal*
Application event totals		17.42	4.78	20.05	325.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	15.10	4.15	17.38	84.62	93,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	217.11	1,452,132.00 *gal*
Application event totals		15.10	4.15	17.38	301.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	23.23	6.38	26.74	130.18	144,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	180.64	1,208,161.00 *gal*
Application event totals		23.23	6.38	26.74	310.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.56	7.02	29.41	143.20	158,400.00 *gal*
Canal	Surface water	0.00	0.00	0.00	164.07	1,097,347.00 *gal*
Application event totals		25.56	7.02	29.41	307.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	24.39	6.70	28.07	136.69	151,200.00 *gal*
Canal	Surface water	0.00	0.00	0.00	152.83	1,022,146.00 *gal*
Application event totals		24.39	6.70	28.07	289.52	

BUNGARD - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.56	7.02	29.41	143.20	158,400.00 *gal*
Canal	Surface water	0.00	0.00	0.00	149.16	997,623.00 *gal*
Application event totals		25.56	7.02	29.41	292.36	

ELMQUIST - 11/08/2022: Oats, silage-soft dough

Field name: ELMQUIST

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.57	6.20	54.68	518.73	288,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	301.58	4,034,160.00 *gal*
Application event totals		46.57	6.20	54.68	820.31	

02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.90	2.46	56.43	353.54	165,600.00 *gal*
Application event totals		31.90	2.46	56.43	353.54	

03/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.77	3.54	80.96	507.26	237,600.00 *gal*
Application event totals		45.77	3.54	80.96	507.26	

ELMQUIST - 11/08/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.94	4.55	104.27	653.29	306,000.00 *gal*
Application event totals		58.94	4.55	104.27	653.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	18.03	1.39	31.89	199.83	93,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	240.40	3,215,736.00 *gal*
Application event totals		18.03	1.39	31.89	440.23	

ELMQUIST - 06/01/2023: Corn, silage

Field name: ELMQUIST

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	280.07	3,746,376.00 *gal*
Application event totals		0.00	0.00	0.00	280.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	90.15	6.96	159.47	999.15	468,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	254.95	3,410,376.00 *gal*
Application event totals		90.15	6.96	159.47	1,254.10	

ELMQUIST - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Subsurface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.43	7.26	30.41	148.08	327,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	247.61	3,312,192.00 *gal*
Application event totals		26.43	7.26	30.41	395.69	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.78	8.45	35.43	172.49	381,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	224.36	3,001,200.00 *gal*
Application event totals		30.78	8.45	35.43	396.85	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	19.17	5.26	22.06	107.40	237,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	217.08	2,903,760.00 *gal*
Application event totals		19.17	5.26	22.06	324.48	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.21	13.24	55.48	270.13	597,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	196.32	2,626,117.00 *gal*
Application event totals		48.21	13.24	55.48	466.45	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	23.23	6.38	26.74	130.18	288,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	149.59	2,000,943.00 *gal*
Application event totals		23.23	6.38	26.74	279.77	

ELMQUIST - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.49	8.37	35.09	170.86	378,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	133.78	1,789,482.00 *gal*
Application event totals		30.49	8.37	35.09	304.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	24.97	6.86	28.74	139.95	309,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	143.99	1,926,114.00 *gal*
Application event totals		24.97	6.86	28.74	283.94	

MCCULLOUGH - 11/08/2022: Oats, silage-soft dough

Field name: MCCULLOUGH

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	301.27	3,358,300.00 *gal*
Application event totals		0.00	0.00	0.00	301.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.61	3.21	73.60	461.14	180,000.00 *gal*
Application event totals		41.61	3.21	73.60	461.14	

MCCULLOUGH - 11/08/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	22.47	1.74	39.75	249.02	97,200.00 *gal*
Application event totals		22.47	1.74	39.75	249.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.92	4.24	97.16	608.71	237,600.00 *gal*
Application event totals		54.92	4.24	97.16	608.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.93	3.86	88.32	553.37	216,000.00 *gal*
Application event totals		49.93	3.86	88.32	553.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	21.64	1.67	38.27	239.80	93,600.00 *gal*
Application event totals		21.64	1.67	38.27	239.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.29	2.57	58.88	368.92	144,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	253.34	2,824,040.00 *gal*
Application event totals		33.29	2.57	58.88	622.26	

MCCULLOUGH - 06/01/2023: Corn, silage

Field name: MCCULLOUGH

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

MCCULLOUGH - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	272.32	3,035,560.00 *gal*
Application event totals		0.00	0.00	0.00	272.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	258.36	2,879,960.00 *gal*
Application event totals		0.00	0.00	0.00	258.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.61	12.25	51.34	249.95	460,800.00 *gal*
Canal	Surface water	0.00	0.00	0.00	244.23	2,722,500.00 *gal*
Application event totals		44.61	12.25	51.34	494.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.18	13.78	57.75	281.19	518,400.00 *gal*
Canal	Surface water	0.00	0.00	0.00	232.46	2,591,300.00 *gal*
Application event totals		50.18	13.78	57.75	513.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.88	13.97	58.55	285.10	525,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	218.98	2,441,000.00 *gal*
Application event totals		50.88	13.97	58.55	504.08	

MCCULLOUGH - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.23	14.07	58.96	287.05	529,200.00 *gal*
Canal	Surface water	0.00	0.00	0.00	196.25	2,187,613.00 *gal*
Application event totals		51.23	14.07	58.96	483.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	23.70	6.51	27.27	132.79	244,800.00 *gal*
Canal	Surface water	0.00	0.00	0.00	162.16	1,807,631.00 *gal*
Application event totals		23.70	6.51	27.27	294.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.97	8.23	34.49	167.93	309,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	145.95	1,626,891.00 *gal*
Application event totals		29.97	8.23	34.49	313.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.85	9.57	40.11	195.27	360,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	139.19	1,551,613.00 *gal*
Application event totals		34.85	9.57	40.11	334.47	

NEWMAN - 11/12/2022: Oats, silage-soft dough

Field name: NEWMAN

Crop: Oats, silage-soft dough

Plant date: 11/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

NEWMAN - 11/12/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	317.83	4,782,915.00 *gal*
Application event totals		0.00	0.00	0.00	317.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.74	2.14	49.07	307.43	162,000.00 *gal*
Application event totals		27.74	2.14	49.07	307.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.12	2.10	47.98	300.60	158,400.00 *gal*
Application event totals		27.12	2.10	47.98	300.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.83	3.00	68.70	430.40	226,800.00 *gal*
Application event totals		38.83	3.00	68.70	430.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.99	2.86	65.42	409.91	216,000.00 *gal*
Application event totals		36.99	2.86	65.42	409.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.68	3.14	71.97	450.90	237,600.00 *gal*
Application event totals		40.68	3.14	71.97	450.90	

NEWMAN - 11/12/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.31	3.81	87.23	546.54	288,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	241.33	3,631,716.00 *gal*
Application event totals		49.31	3.81	87.23	787.87	

NEWMAN - 06/01/2023: Corn, silage

Field name: NEWMAN

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	110.96	8.57	196.27	1,229.72	648,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	276.68	4,163,724.00 *gal*
Application event totals		110.96	8.57	196.27	1,506.40	

06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	73.97	5.71	130.85	819.81	432,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	259.50	3,905,073.00 *gal*
Application event totals		73.97	5.71	130.85	1,079.31	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	239.62	3,605,904.00 *gal*
Application event totals		0.00	0.00	0.00	239.62	

NEWMAN - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	37.17	10.21	42.78	208.29	518,400.00 *gal*
Canal	Surface water	0.00	0.00	0.00	228.87	3,444,147.00 *gal*
Application event totals		37.17	10.21	42.78	437.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	217.51	3,273,183.00 *gal*
Application event totals		0.00	0.00	0.00	217.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.36	7.51	31.49	153.33	381,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	191.49	2,881,700.00 *gal*
Application event totals		27.36	7.51	31.49	344.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	21.68	5.95	24.95	121.50	302,400.00 *gal*
Canal	Surface water	0.00	0.00	0.00	162.55	2,446,116.00 *gal*
Application event totals		21.68	5.95	24.95	284.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.85	7.37	30.90	150.43	374,400.00 *gal*
Canal	Surface water	0.00	0.00	0.00	149.82	2,254,663.00 *gal*
Application event totals		26.85	7.37	30.90	300.26	

NEWMAN - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	133.82	2,013,754.00 *gal*
Application event totals		0.00	0.00	0.00	133.82	

REDDING EAST - 11/08/2022: Oats, silage-soft dough

Field name: REDDING EAST

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.80	5.17	45.56	432.27	90,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	304.83	1,529,082.00 *gal*
Application event totals		38.80	5.17	45.56	737.10	

03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.99	2.86	65.42	409.91	72,000.00 *gal*
Application event totals		36.99	2.86	65.42	409.91	

04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.83	3.00	68.70	430.40	75,600.00 *gal*
Application event totals		38.83	3.00	68.70	430.40	

REDDING EAST - 11/08/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	92.46	7.14	163.56	1,024.77	180,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	251.57	1,261,935.00 *gal*
Application event totals		92.46	7.14	163.56	1,276.34	

REDDING EAST - 06/01/2023: Corn, silage

Field name: REDDING EAST

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	110.96	8.57	196.27	1,229.72	216,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	252.38	1,265,975.00 *gal*
Application event totals		110.96	8.57	196.27	1,482.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	73.97	5.71	130.85	819.81	144,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	241.16	1,209,725.00 *gal*
Application event totals		73.97	5.71	130.85	1,060.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	23.23	6.38	26.74	130.18	108,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	250.51	1,256,634.00 *gal*
Application event totals		23.23	6.38	26.74	380.70	

REDDING EAST - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	21.68	5.95	24.95	121.50	100,800.00 *gal*
Canal	Surface water	0.00	0.00	0.00	228.91	1,148,256.00 *gal*
Application event totals		21.68	5.95	24.95	350.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	4.65	1.28	5.35	26.04	21,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	217.68	1,091,934.00 *gal*
Application event totals		4.65	1.28	5.35	243.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	15.49	4.25	17.82	86.79	72,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	197.44	990,423.00 *gal*
Application event totals		15.49	4.25	17.82	284.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	20.13	5.53	23.17	112.82	93,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	135.05	677,463.00 *gal*
Application event totals		20.13	5.53	23.17	247.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	4.65	1.28	5.35	26.04	21,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	175.32	879,460.00 *gal*
Application event totals		4.65	1.28	5.35	201.36	

REDDING EAST - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	15.49	4.25	17.82	86.79	72,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	153.54	770,216.00 *gal*
Application event totals		15.49	4.25	17.82	240.33	

REDDING WEST - 11/08/2022: Oats, silage-soft dough

Field name: REDDING WEST

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.70	4.75	41.92	397.69	165,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	290.93	2,918,790.00 *gal*
Application event totals		35.70	4.75	41.92	688.62	

01/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.99	2.86	65.42	409.91	144,000.00 *gal*
Application event totals		36.99	2.86	65.42	409.91	

02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.53	3.29	75.24	471.39	165,600.00 *gal*
Application event totals		42.53	3.29	75.24	471.39	

REDDING WEST - 11/08/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	24.04	1.86	42.53	266.44	93,600.00 *gal*
Application event totals		24.04	1.86	42.53	266.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	77.67	6.00	137.39	860.80	302,400.00 *gal*
Application event totals		77.67	6.00	137.39	860.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	237.03	2,377,962.00 *gal*
Application event totals		0.00	0.00	0.00	237.03	

REDDING WEST - 06/01/2023: Corn, silage

Field name: REDDING WEST

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.72	5.00	114.49	717.34	252,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	279.98	2,808,900.00 *gal*
Application event totals		64.72	5.00	114.49	997.32	

REDDING WEST - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	24.04	1.86	42.53	266.44	93,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	264.02	2,648,790.00 *gal*
Application event totals		24.04	1.86	42.53	530.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	15.49	4.25	17.82	86.79	144,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	235.68	2,364,470.00 *gal*
Application event totals		15.49	4.25	17.82	322.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	24.78	6.80	28.52	138.86	230,400.00 *gal*
Canal	Surface water	0.00	0.00	0.00	167.89	1,684,392.00 *gal*
Application event totals		24.78	6.80	28.52	306.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.72	10.63	44.56	216.97	360,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	217.50	2,182,104.00 *gal*
Application event totals		38.72	10.63	44.56	434.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.17	6.91	28.97	141.03	234,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	196.52	1,971,603.00 *gal*
Application event totals		25.17	6.91	28.97	337.55	

REDDING WEST - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.85	9.57	40.11	195.27	324,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	176.92	1,774,897.00 *gal*
Application event totals		34.85	9.57	40.11	372.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.98	8.51	35.65	173.58	288,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	150.33	1,508,177.00 *gal*
Application event totals		30.98	8.51	35.65	323.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.72	10.63	44.56	216.97	360,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	141.94	1,424,016.00 *gal*
Application event totals		38.72	10.63	44.56	358.91	

TELL - 11/14/2022: Oats, silage-soft dough

Field name: TELL

Crop: Oats, silage-soft dough

Plant date: 11/14/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	320.66	4,468,025.00 *gal*
Application event totals		0.00	0.00	0.00	320.66	

TELL - 11/14/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.60	3.60	82.44	516.48	252,000.00 *gal*
Application event totals		46.60	3.60	82.44	516.48	

02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.62	2.37	54.17	339.40	165,600.00 *gal*
Application event totals		30.62	2.37	54.17	339.40	

03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.63	2.06	47.11	295.13	144,000.00 *gal*
Application event totals		26.63	2.06	47.11	295.13	

03/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.59	4.22	96.57	605.02	295,200.00 *gal*
Application event totals		54.59	4.22	96.57	605.02	

04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.96	2.47	56.53	354.16	172,800.00 *gal*
Application event totals		31.96	2.47	56.53	354.16	

04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	253.34	3,530,050.00 *gal*
Application event totals		0.00	0.00	0.00	253.34	

TELL - 06/01/2023: Corn, silage

TELL - 06/01/2023: Corn, silage

Field name: TELL

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.92	4.63	105.99	664.05	324,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	273.05	3,804,700.00 *gal*
Application event totals		59.92	4.63	105.99	937.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	83.88	6.48	148.38	929.67	453,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	242.73	3,382,200.00 *gal*
Application event totals		83.88	6.48	148.38	1,172.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	23.98	6.58	27.59	134.35	309,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	233.22	3,249,750.00 *gal*
Application event totals		23.98	6.58	27.59	367.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	223.39	3,112,700.00 *gal*
Application event totals		0.00	0.00	0.00	223.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.57	9.49	39.79	193.71	446,400.00 *gal*
Canal	Surface water	0.00	0.00	0.00	215.36	3,000,825.00 *gal*
Application event totals		34.57	9.49	39.79	409.07	

TELL - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	16.73	4.59	19.25	93.73	216,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	210.34	2,930,860.00 *gal*
Application event totals		16.73	4.59	19.25	304.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.13	9.64	40.43	196.84	453,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	195.50	2,724,113.00 *gal*
Application event totals		35.13	9.64	40.43	392.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.45	9.19	38.50	187.46	432,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	186.00	2,591,761.00 *gal*
Application event totals		33.45	9.19	38.50	373.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	160.88	2,241,703.00 *gal*
Application event totals		0.00	0.00	0.00	160.88	

WRIGHTS - 11/11/2022: Oats, silage-soft dough

Field name: WRIGHTS

Crop: Oats, silage-soft dough

Plant date: 11/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

WRIGHTS - 11/11/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/23/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	302.34	1,179,584.00 *gal*
Application event totals		0.00	0.00	0.00	302.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.57	5.14	117.76	737.83	100,800.00 *gal*
Application event totals		66.57	5.14	117.76	737.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	14.27	1.10	25.24	158.11	21,600.00 *gal*
Application event totals		14.27	1.10	25.24	158.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.95	5.33	121.97	764.18	104,400.00 *gal*
Application event totals		68.95	5.33	121.97	764.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.55	3.67	84.12	527.02	72,000.00 *gal*
Application event totals		47.55	3.67	84.12	527.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	14.27	1.10	25.24	158.11	21,600.00 *gal*
Application event totals		14.27	1.10	25.24	158.11	

WRIGHTS - 11/11/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	254.88	994,406.00 *gal*
Application event totals		0.00	0.00	0.00	254.88	

WRIGHTS - 06/01/2023: Corn, silage

Field name: WRIGHTS

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	95.10	7.35	168.24	1,054.05	144,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	272.91	1,064,756.00 *gal*
Application event totals		95.10	7.35	168.24	1,326.95	

06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.55	3.67	84.12	527.02	72,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	262.41	1,023,813.00 *gal*
Application event totals		47.55	3.67	84.12	789.44	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	5.97	1.64	6.88	33.48	21,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	248.42	969,206.00 *gal*
Application event totals		5.97	1.64	6.88	281.89	

WRIGHTS - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.88	7.65	32.08	156.22	100,800.00 *gal*
Canal	Surface water	0.00	0.00	0.00	216.22	843,584.00 *gal*
Application event totals		27.88	7.65	32.08	372.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.83	10.94	45.84	223.17	144,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	208.29	812,630.00 *gal*
Application event totals		39.83	10.94	45.84	431.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.87	8.20	34.38	167.38	108,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	179.52	700,413.00 *gal*
Application event totals		29.87	8.20	34.38	346.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	19.91	5.47	22.92	111.58	72,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	164.61	642,216.00 *gal*
Application event totals		19.91	5.47	22.92	276.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	5.97	1.64	6.88	33.48	21,600.00 *gal*
Canal	Surface water	0.00	0.00	0.00	123.06	480,130.00 *gal*
Application event totals		5.97	1.64	6.88	156.54	

WRIGHTS - 06/01/2023: Corn, silage

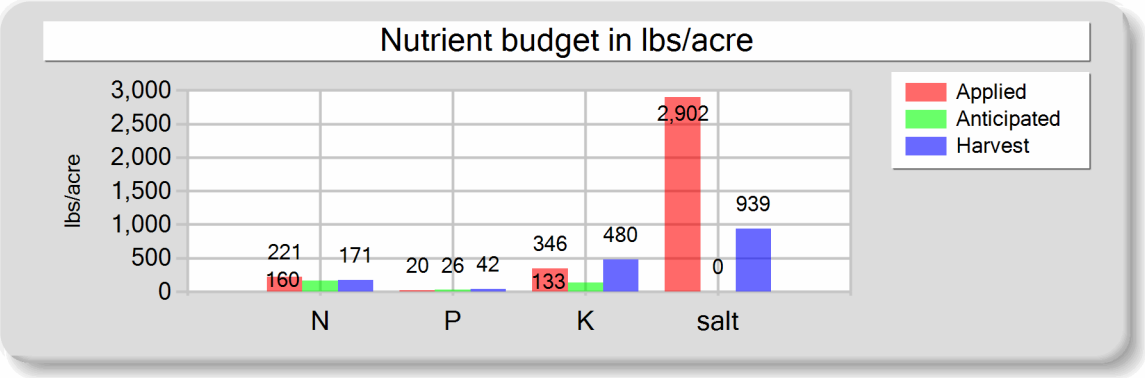
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	19.91	5.47	22.92	111.58	72,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	102.59	400,266.00 *gal*
Application event totals		19.91	5.47	22.92	214.18	

B. NUTRIENT BUDGET

BLEVINS - 11/13/2022: Oats, silage-soft dough

Field name: BLEVINS | Crop: Oats, silage-soft dough | Plant date: 11/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	213.81	19.55	345.97	2,372.79
Fresh water	0.00	0.00	0.00	529.39
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	220.81	19.55	345.97	2,902.18
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	171.00	42.00	480.00	939.12
Nutrient balance	49.81	-22.45	-134.03	1,963.06
Applied to removed ratio	1.29	0.47	0.72	3.09

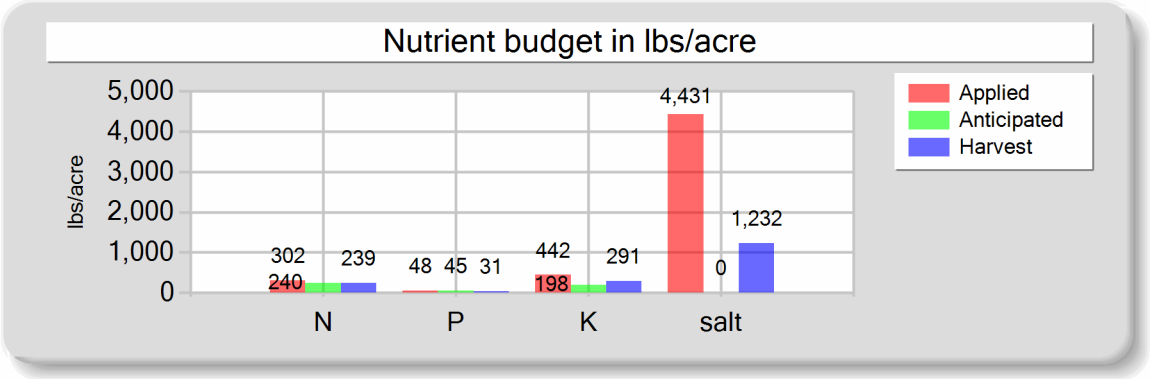
Fresh water applied	
5,016,071.00	*gallons*
184.72	*acre-inches*
10.87	*inches/acre*

Process wastewater applied	
824,400.00	*gallons*
30.36	*acre-inches*
1.79	*inches/acre*

Total harvests for the crop	
1	*harvests*

BLEVINS - 06/01/2023: Corn, silage

Field name: BLEVINS Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	295.42	48.47	442.25	2,561.98
Fresh water	0.00	0.00	0.00	1,869.02
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	302.42	48.47	442.25	4,431.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	239.20	31.20	291.20	1,232.40
Nutrient balance	63.22	17.27	151.05	3,198.60
Applied to removed ratio	1.26	1.55	1.52	3.60

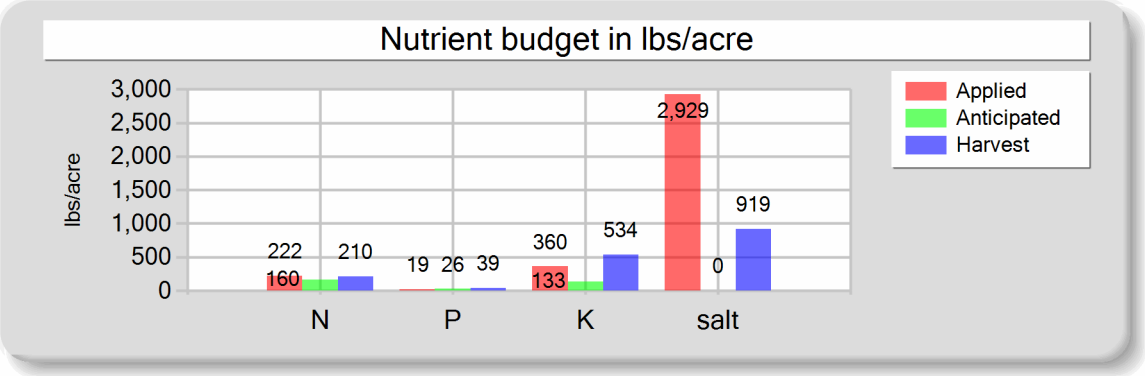
Fresh water applied	
17,709,194.00	*gallons*
652.17	*acre-inches*
38.36	*inches/acre*

Process wastewater applied	
1,749,600.00	*gallons*
64.43	*acre-inches*
3.79	*inches/acre*

Total harvests for the crop	
1	*harvests*

BUNGARD - 11/13/2022: Oats, silage-soft dough

Field name: BUNGARD Crop: Oats, silage-soft dough Plant date: 11/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	215.23	18.58	359.95	2,387.34
Fresh water	0.00	0.00	0.00	541.76
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	222.23	18.58	359.95	2,929.10
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	210.17	38.80	533.50	918.59
Nutrient balance	12.06	-20.22	-173.55	2,010.51
Applied to removed ratio	1.06	0.48	0.67	3.19

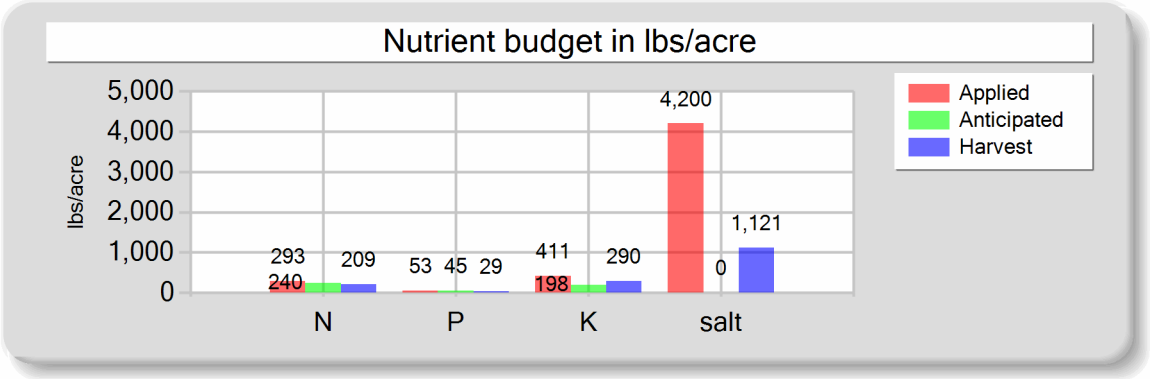
Fresh water applied	
3,623,460.00	*gallons*
133.44	*acre-inches*
11.12	*inches/acre*

Process wastewater applied	
576,000.00	*gallons*
21.21	*acre-inches*
1.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

BUNGARD - 06/01/2023: Corn, silage

Field name: BUNGARD Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	286.26	52.60	410.88	2,326.00
Fresh water	0.00	0.00	0.00	1,874.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	293.26	52.60	410.88	4,200.29
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	209.44	28.56	290.36	1,120.98
Nutrient balance	83.82	24.04	120.52	3,079.31
Applied to removed ratio	1.40	1.84	1.42	3.75

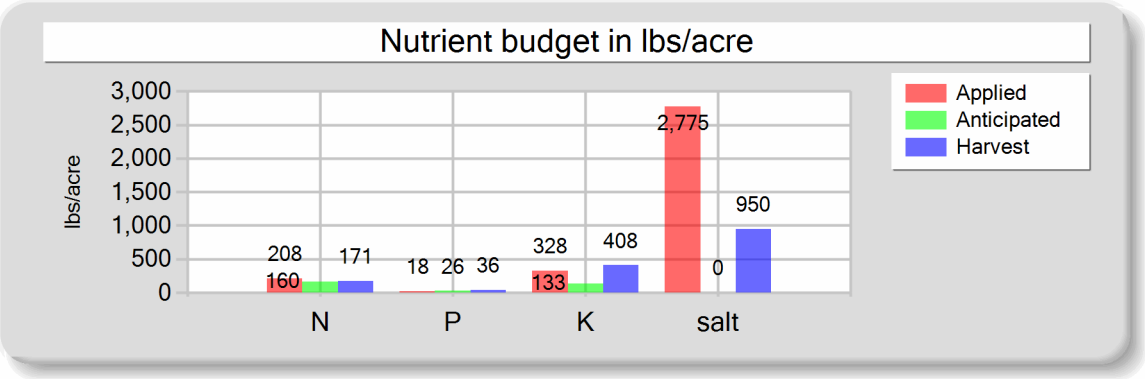
Fresh water applied	
12,535,797.00	*gallons*
461.65	*acre-inches*
38.47	*inches/acre*

Process wastewater applied	
1,299,600.00	*gallons*
47.86	*acre-inches*
3.99	*inches/acre*

Total harvests for the crop	
1	*harvests*

ELMQUIST - 11/08/2022: Oats, silage-soft dough

Field name: ELMQUIST Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.21	18.15	328.23	2,232.65
Fresh water	0.00	0.00	0.00	541.98
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	208.21	18.15	328.23	2,774.63
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	170.50	36.17	408.17	950.07
Nutrient balance	37.71	-18.02	-79.93	1,824.56
Applied to removed ratio	1.22	0.50	0.80	2.92

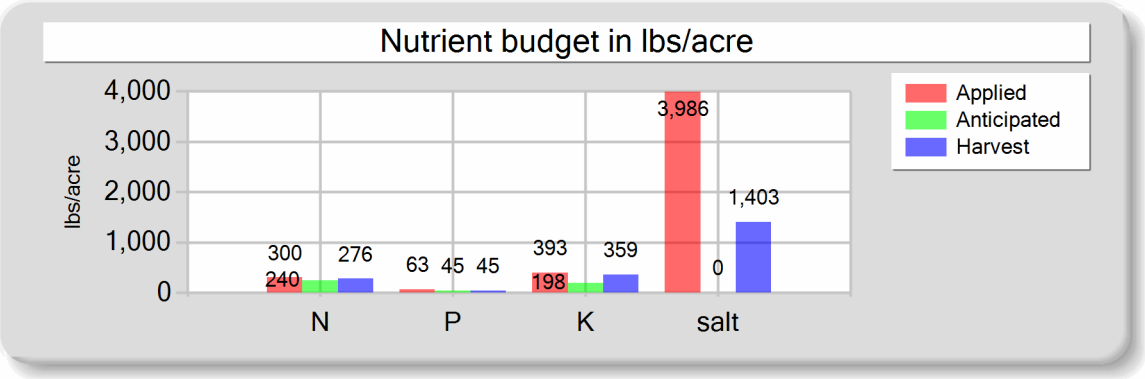
Fresh water applied	
7,249,896.00	*gallons*
266.99	*acre-inches*
11.12	*inches/acre*

Process wastewater applied	
1,090,800.00	*gallons*
40.17	*acre-inches*
1.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

ELMQUIST - 06/01/2023: Corn, silage

Field name: ELMQUIST Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	293.44	62.78	393.43	2,138.24
Fresh water	0.00	0.00	0.00	1,847.74
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	300.44	62.78	393.43	3,985.98
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	275.92	44.92	359.33	1,403.20
Nutrient balance	24.52	17.86	34.09	2,582.79
Applied to removed ratio	1.09	1.40	1.09	2.84

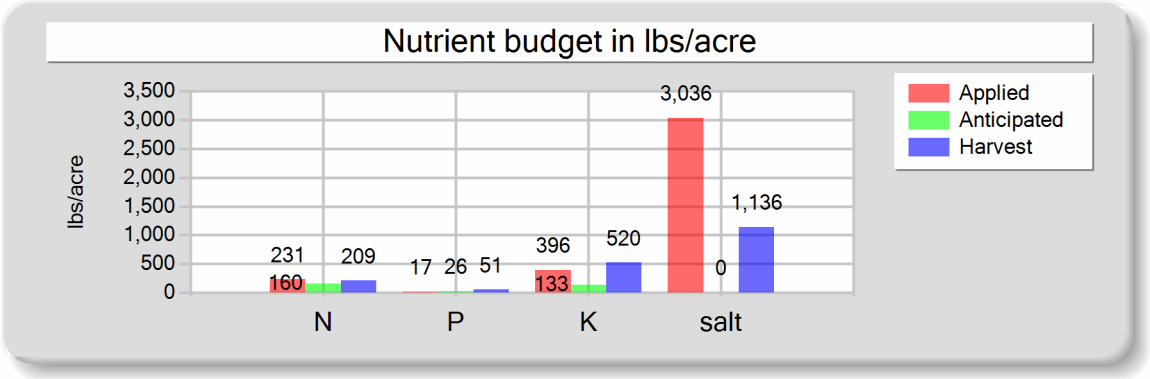
Fresh water applied	
24,716,560.00	*gallons*
910.23	*acre-inches*
37.93	*inches/acre*

Process wastewater applied	
2,988,000.00	*gallons*
110.04	*acre-inches*
4.58	*inches/acre*

Total harvests for the crop	
1	*harvests*

MCCULLOUGH - 11/08/2022: Oats, silage-soft dough

Field name: MCCULLOUGH Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	223.85	17.29	395.98	2,480.96
Fresh water	0.00	0.00	0.00	554.61
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	230.85	17.29	395.98	3,035.57
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	208.65	51.36	520.02	1,136.34
Nutrient balance	22.20	-34.07	-124.04	1,899.23
Applied to removed ratio	1.11	0.34	0.76	2.67

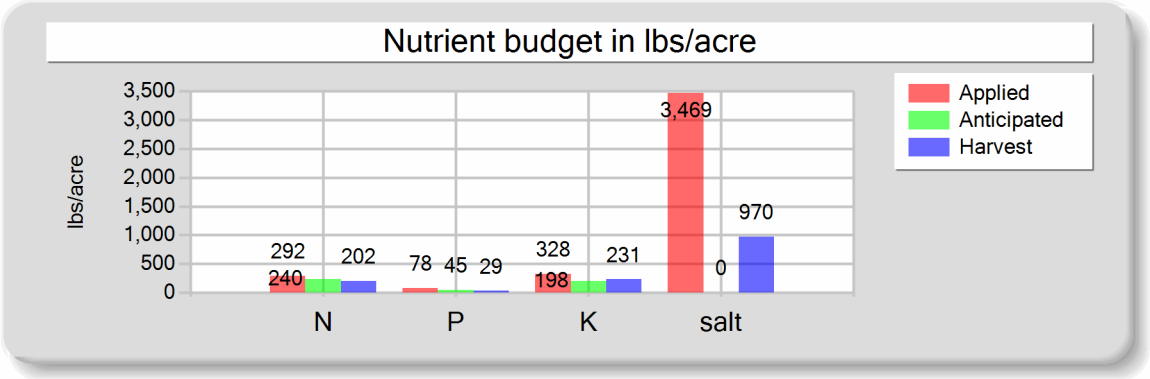
Fresh water applied	
6,182,340.00	*gallons*
227.67	*acre-inches*
11.38	*inches/acre*

Process wastewater applied	
968,400.00	*gallons*
35.66	*acre-inches*
1.78	*inches/acre*

Total harvests for the crop	
1	*harvests*

MCCULLOUGH - 06/01/2023: Corn, silage

Field name: MCCULLOUGH Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	285.41	78.37	328.47	1,599.29
Fresh water	0.00	0.00	0.00	1,869.90
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	292.41	78.37	328.47	3,469.18
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	202.44	28.92	231.36	970.17
Nutrient balance	89.97	49.45	97.11	2,499.01
Applied to removed ratio	1.44	2.71	1.42	3.58

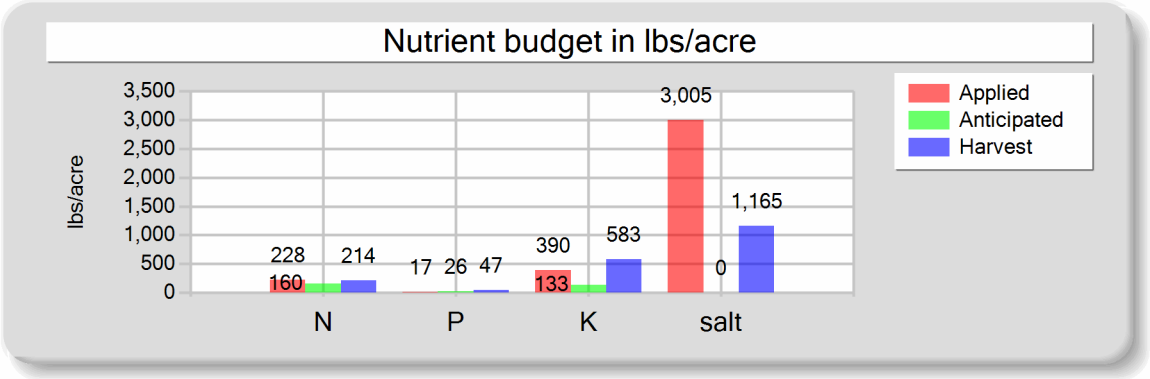
Fresh water applied	
20,844,068.00	*gallons*
767.62	*acre-inches*
38.38	*inches/acre*

Process wastewater applied	
2,948,400.00	*gallons*
108.58	*acre-inches*
5.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

NEWMAN - 11/12/2022: Oats, silage-soft dough

Field name: NEWMAN Crop: Oats, silage-soft dough Plant date: 11/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	220.68	17.05	390.37	2,445.77
Fresh water	0.00	0.00	0.00	559.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	227.68	17.05	390.37	3,004.93
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	214.28	46.87	582.58	1,165.16
Nutrient balance	13.40	-29.83	-192.21	1,839.78
Applied to removed ratio	1.06	0.36	0.67	2.58

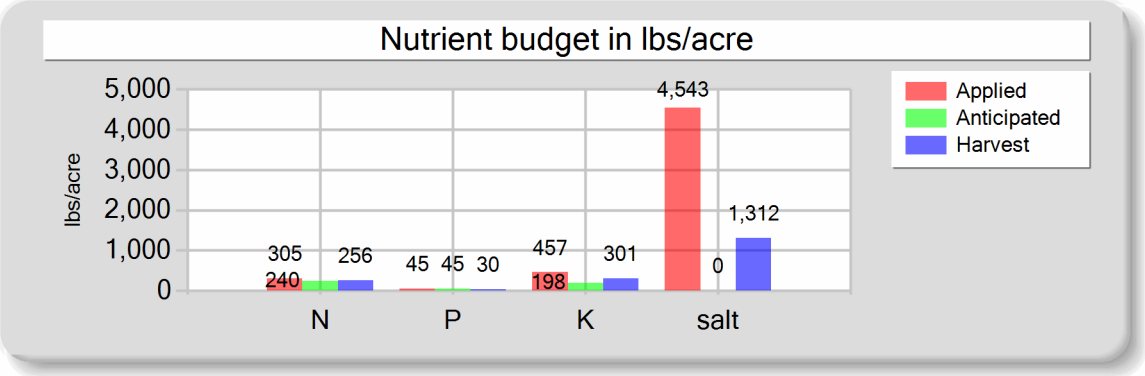
Fresh water applied	
8,414,631.00	*gallons*
309.88	*acre-inches*
11.48	*inches/acre*

Process wastewater applied	
1,288,800.00	*gallons*
47.46	*acre-inches*
1.76	*inches/acre*

Total harvests for the crop	
1	*harvests*

NEWMAN - 06/01/2023: Corn, silage

Field name: NEWMAN Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	297.99	45.33	457.25	2,683.08
Fresh water	0.00	0.00	0.00	1,859.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	304.99	45.33	457.25	4,542.93
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	256.13	30.13	301.33	1,311.52
Nutrient balance	48.86	15.20	155.91	3,231.41
Applied to removed ratio	1.19	1.50	1.52	3.46

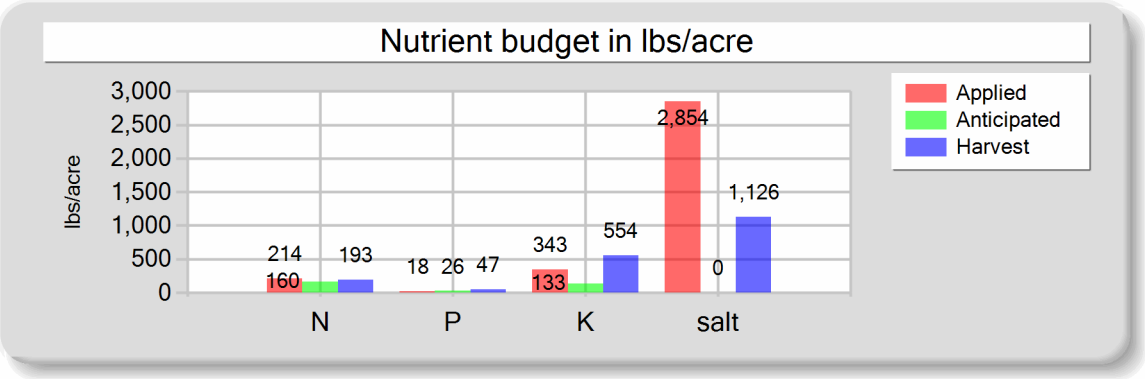
Fresh water applied	
27,988,264.00	*gallons*
1,030.71	*acre-inches*
38.17	*inches/acre*

Process wastewater applied	
2,656,800.00	*gallons*
97.84	*acre-inches*
3.62	*inches/acre*

Total harvests for the crop	
1	*harvests*

REDDING EAST - 11/08/2022: Oats, silage-soft dough

Field name: REDDING EAST Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	207.09	18.17	343.25	2,297.35
Fresh water	0.00	0.00	0.00	556.40
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	214.09	18.17	343.25	2,853.74
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	193.16	47.38	553.96	1,126.50
Nutrient balance	20.93	-29.21	-210.71	1,727.24
Applied to removed ratio	1.11	0.38	0.62	2.53

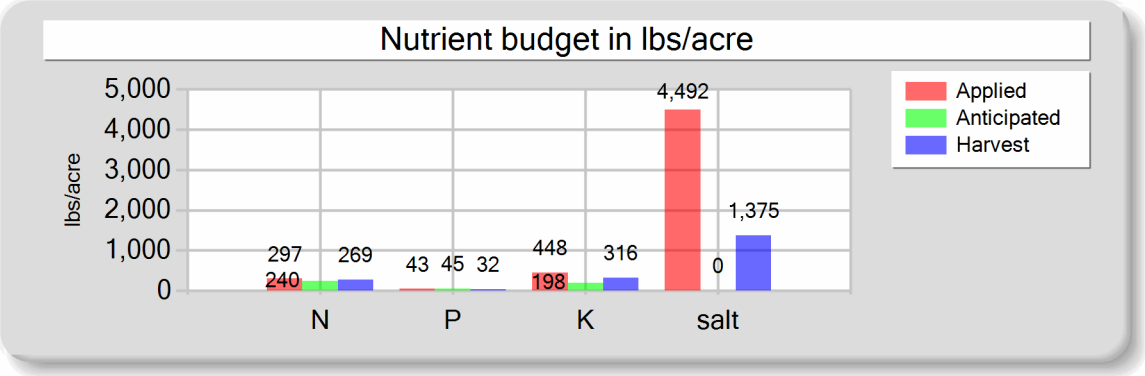
Fresh water applied	
2,791,017.00	*gallons*
102.78	*acre-inches*
11.42	*inches/acre*

Process wastewater applied	
417,600.00	*gallons*
15.38	*acre-inches*
1.71	*inches/acre*

Total harvests for the crop	
1	*harvests*

REDDING EAST - 06/01/2023: Corn, silage

Field name: REDDING EAST Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	290.25	43.20	448.33	2,639.69
Fresh water	0.00	0.00	0.00	1,852.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.25	43.20	448.33	4,491.69
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	268.60	31.60	316.00	1,375.36
Nutrient balance	28.65	11.60	132.33	3,116.34
Applied to removed ratio	1.11	1.37	1.42	3.27

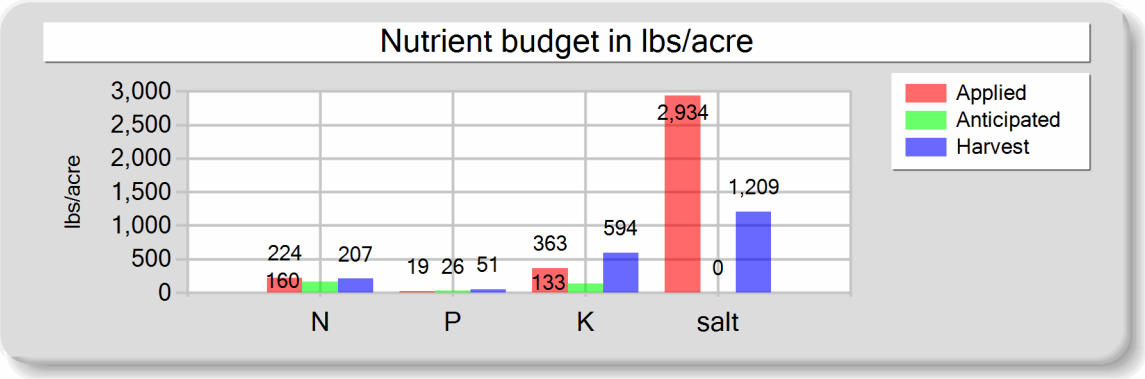
Fresh water applied	
9,290,086.00	*gallons*
342.12	*acre-inches*
38.01	*inches/acre*

Process wastewater applied	
849,600.00	*gallons*
31.29	*acre-inches*
3.48	*inches/acre*

Total harvests for the crop	
1	*harvests*

REDDING WEST - 11/08/2022: Oats, silage-soft dough

Field name: REDDING WEST Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	216.93	18.75	362.50	2,406.23
Fresh water	0.00	0.00	0.00	527.96
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	223.93	18.75	362.50	2,934.19
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	207.29	50.84	594.49	1,208.92
Nutrient balance	16.64	-32.09	-231.99	1,725.27
Applied to removed ratio	1.08	0.37	0.61	2.43

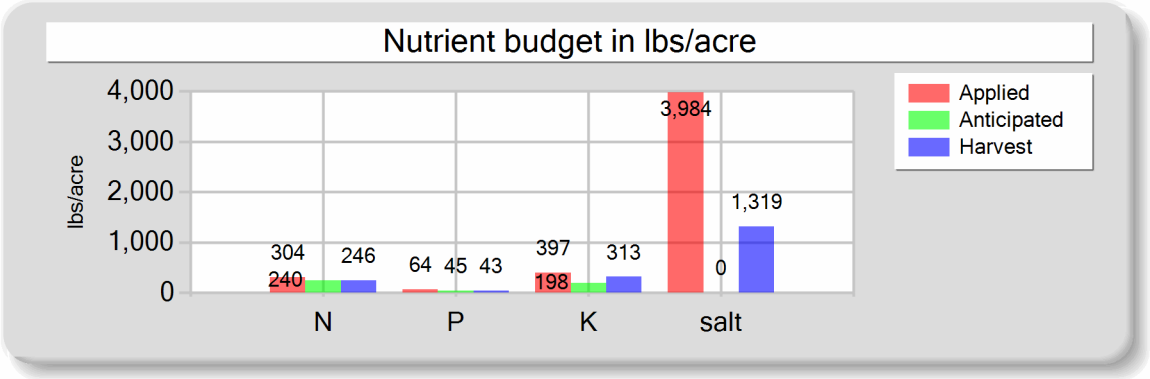
Fresh water applied	
5,296,752.00	*gallons*
195.06	*acre-inches*
10.84	*inches/acre*

Process wastewater applied	
871,200.00	*gallons*
32.08	*acre-inches*
1.78	*inches/acre*

Total harvests for the crop	
1	*harvests*

REDDING WEST - 06/01/2023: Corn, silage

Field name: REDDING WEST Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	297.47	64.16	397.21	2,153.24
Fresh water	0.00	0.00	0.00	1,830.79
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	304.47	64.16	397.21	3,984.03
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	245.78	43.01	313.37	1,318.72
Nutrient balance	58.69	21.15	83.84	2,665.31
Applied to removed ratio	1.24	1.49	1.27	3.02

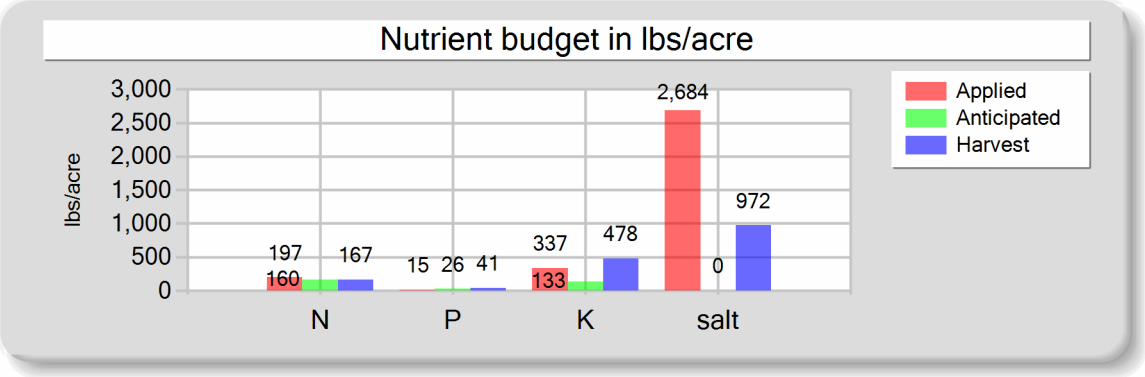
Fresh water applied	
18,367,349.00	*gallons*
676.41	*acre-inches*
37.58	*inches/acre*

Process wastewater applied	
2,286,000.00	*gallons*
84.19	*acre-inches*
4.68	*inches/acre*

Total harvests for the crop	
1	*harvests*

TELL - 11/14/2022: Oats, silage-soft dough

Field name: TELL Crop: Oats, silage-soft dough Plant date: 11/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	190.40	14.71	336.81	2,110.20
Fresh water	0.00	0.00	0.00	574.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	197.40	14.71	336.81	2,684.20
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	166.63	40.87	477.89	971.81
Nutrient balance	30.77	-26.16	-141.08	1,712.39
Applied to removed ratio	1.18	0.36	0.70	2.76

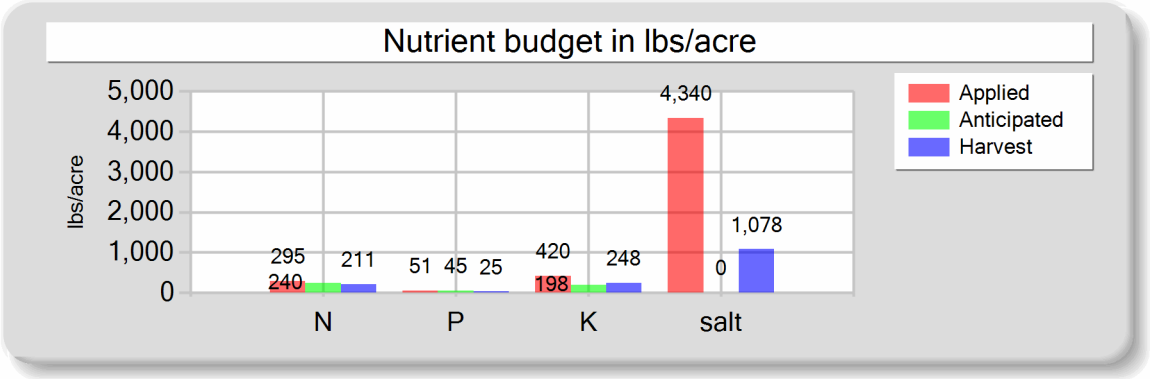
Fresh water applied	
7,998,075.00	*gallons*
294.54	*acre-inches*
11.78	*inches/acre*

Process wastewater applied	
1,029,600.00	*gallons*
37.92	*acre-inches*
1.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

TELL - 06/01/2023: Corn, silage

Field name: TELL | Crop: Corn, silage | Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	287.65	50.61	419.93	2,399.80
Fresh water	0.00	0.00	0.00	1,940.48
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	294.65	50.61	419.93	4,340.28
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	210.53	24.77	247.68	1,078.00
Nutrient balance	84.13	25.84	172.25	3,262.28
Applied to removed ratio	1.40	2.04	1.70	4.03

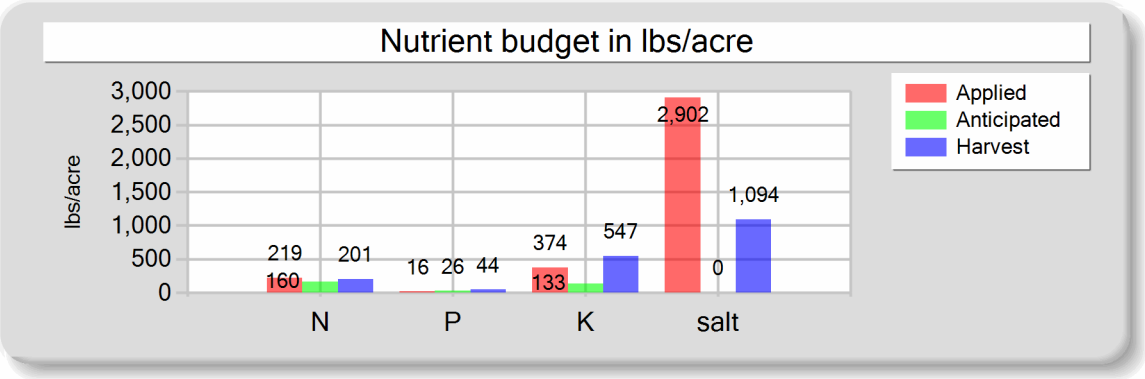
Fresh water applied	
27,038,612.00	*gallons*
995.74	*acre-inches*
39.83	*inches/acre*

Process wastewater applied	
2,635,200.00	*gallons*
97.05	*acre-inches*
3.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

WRIGHTS - 11/11/2022: Oats, silage-soft dough

Field name: WRIGHTS | Crop: Oats, silage-soft dough | Plant date: 11/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	211.61	16.35	374.32	2,345.25
Fresh water	0.00	0.00	0.00	557.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	218.61	16.35	374.32	2,902.47
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	201.14	44.00	546.86	1,093.71
Nutrient balance	17.46	-27.65	-172.53	1,808.75
Applied to removed ratio	1.09	0.37	0.68	2.65

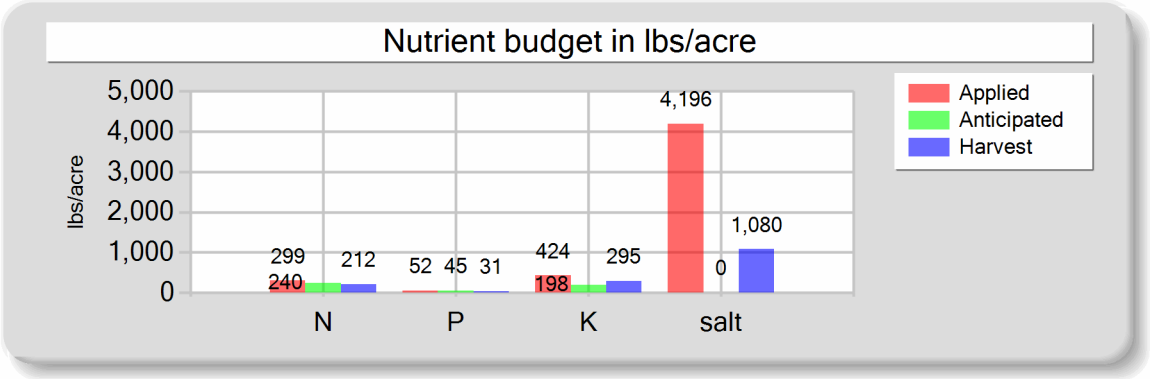
Fresh water applied	
2,173,990.00	*gallons*
80.06	*acre-inches*
11.44	*inches/acre*

Process wastewater applied	
320,400.00	*gallons*
11.80	*acre-inches*
1.69	*inches/acre*

Total harvests for the crop	
1	*harvests*

WRIGHTS - 06/01/2023: Corn, silage

Field name: WRIGHTS Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	292.01	52.03	424.24	2,417.95
Fresh water	0.00	0.00	0.00	1,778.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	299.01	52.03	424.24	4,195.98
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	212.03	31.03	294.77	1,079.69
Nutrient balance	86.98	21.00	129.46	3,116.29
Applied to removed ratio	1.41	1.68	1.44	3.89

Fresh water applied	
6,937,014.00	*gallons*
255.47	*acre-inches*
36.50	*inches/acre*

Process wastewater applied	
756,000.00	*gallons*
27.84	*acre-inches*
3.98	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Spring Corral Solids

Sample and source description: Spring Corral Solids

Sample date: 05/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 38.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,800.00	4,000.00	6,100.00							52.80
DL	100.00	100.00	100.00							0.01

Spring Separated Solids

Sample and source description: Spring Separated Solids

Sample date: 05/01/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 84.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,900.00	6,200.00	6,300.00							21.70
DL	100.00	100.00	100.00							0.01

Fall Corral Solids

Sample and source description: Fall Corral Solids

Sample date: 10/24/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 18.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,500.00	9,700.00	9,800.00							46.70
DL	100.00	100.00	100.00							0.01

Fall Separated Solids

Sample and source description: Fall Separated Solids

Sample date: 10/24/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	2,200.00	4,300.00	10,100.00	3,400.00	1,200.00	2,700.00	650.00		10.80
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

B. PROCESS WASTEWATER ANALYSES

2022 WWSQ4

Sample and source description: 2022 WWSQ4

Sample date: 12/12/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	465.00	311.00	0.00	0.00	61.90	546.00								5,750.00	5,180
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ1

Sample and source description: WWSQ1

Sample date: 03/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,530.00	308.00	0.00	0.00	86.20	508.00								5,270.00	3,860
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ2

Sample and source description: WWSQ2

Sample date: 06/06/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	554.00	386.00	0.00	0.00	42.80	980.00								7,750.00	6,140
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ3

Sample and source description: WWSQ3

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	232.00	120.00	0.00	0.00	63.70	267.00	76.60	51.90	66.10	1,100.00	0.00	77.40	106.00	2,640.00	1,300
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

WWSQ4

Sample and source description: WWSQ4

Sample date: 12/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	624.00	434.00	0.00	0.00	19.10	773.00								7,810.00	7,640
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal

Canal

T.I.D Canal

Sample description: T.I.D Canal

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	215
DL	0.50	0.50	0.50								1.00	10

Irrigation Well

IW # 10

Sample description: IW # 10

Sample date: 09/19/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	30.60	0.00	27.80								1,320.00	740
DL	0.50	0.50	0.50								1.00	10

IW #9

Sample description: IW #9

Sample date: 09/19/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	55.00	0.00	52.20								1,360.00	780
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

BLEVINS - 11/13/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,700.00	1,400.00	16,000.00		10.40
DL	100.00	100.00	100.00		0.01

BLEVINS - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,600.00	600.00	5,600.00		7.90
DL	100.00	100.00	100.00		0.01

BUNGARD - 11/13/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,200.00	1,400.00	18,300.00		9.47
DL	100.00	100.00	100.00		0.01

BUNGARD - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,400.00	600.00	6,100.00		7.50
DL	100.00	100.00	100.00		0.01

ELMQUIST - 11/08/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,600.00	1,400.00	15,800.00		12.30
DL	100.00	100.00	100.00		0.01

ELMQUIST - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,300.00	700.00	5,600.00		7.10
DL	100.00	100.00	100.00		0.01

MCCULLOUGH - 11/08/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,500.00	2,100.00	21,100.00		11.80
DL	100.00	100.00	100.00		0.01

MCCULLOUGH - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	600.00	4,800.00		6.80
DL	100.00	100.00	100.00		0.01

NEWMAN - 11/12/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,800.00	1,800.00	21,100.00		11.60
DL	100.00	100.00	100.00		0.01

NEWMAN - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 74.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	500.00	5,300.00		7.80
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage
Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 73.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,400.00	400.00	4,100.00		7.65
DL	100.00	100.00	100.00		0.01

REDDING EAST - 11/08/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,000.00	1,200.00	18,200.00		11.60
DL	100.00	100.00	100.00		0.01

REDDING EAST - 06/01/2023: Corn, silage

REDDING EAST - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,100.00	600.00	6,000.00		8.06
DL	100.00	100.00	100.00		0.01

REDDING WEST - 11/08/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,000.00	1,200.00	18,200.00		11.60
DL	100.00	100.00	100.00		0.01

REDDING WEST - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	700.00	5,100.00		7.30
DL	100.00	100.00	100.00		0.01

TELL - 11/14/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,600.00	1,400.00	18,200.00		10.50
DL	100.00	100.00	100.00		0.01

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 71.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,300.00	1,300.00	15,200.00		11.00
DL	100.00	100.00	100.00		0.01

TELL - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	700.00	6,200.00		6.60
DL	100.00	100.00	100.00		0.01

TELL - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	600.00	5,700.00		8.90
DL	100.00	100.00	100.00		0.01

WRIGHTS - 11/11/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,100.00	1,600.00	18,200.00		9.87
DL	100.00	100.00	100.00		0.01

WRIGHTS - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	600.00	5,700.00		7.30
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

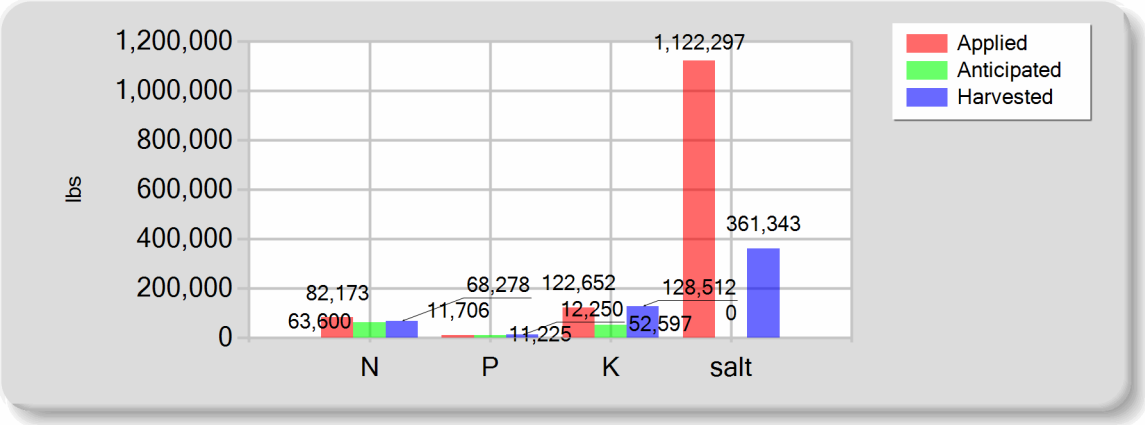
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

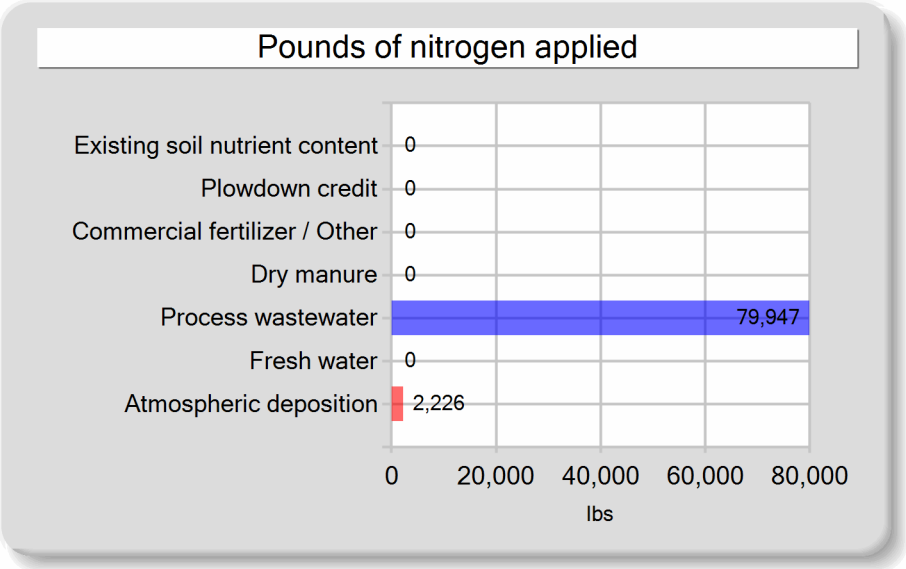
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

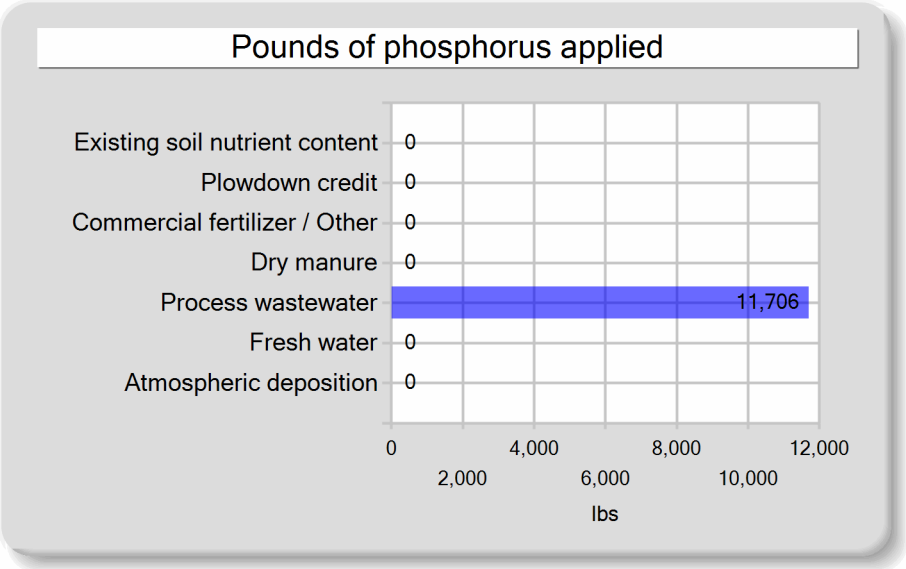
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	79,947.20	11,706.40	122,651.54	738,032.80
Fresh water	0.00	0.00	0.00	384,264.16
Atmospheric deposition	2,226.00	0.00	0.00	0.00
Total nutrients applied	82,173.20	11,706.40	122,651.54	1,122,296.96
Anticipated crop nutrient removal	63,600.00	11,225.40	52,597.20	0.00
Actual crop nutrient removal	68,277.88	12,249.92	128,511.52	361,343.13
Nutrient balance	13,895.32	-543.52	-5,859.98	760,953.83
Applied to removed ratio	1.20	0.96	0.95	3.11

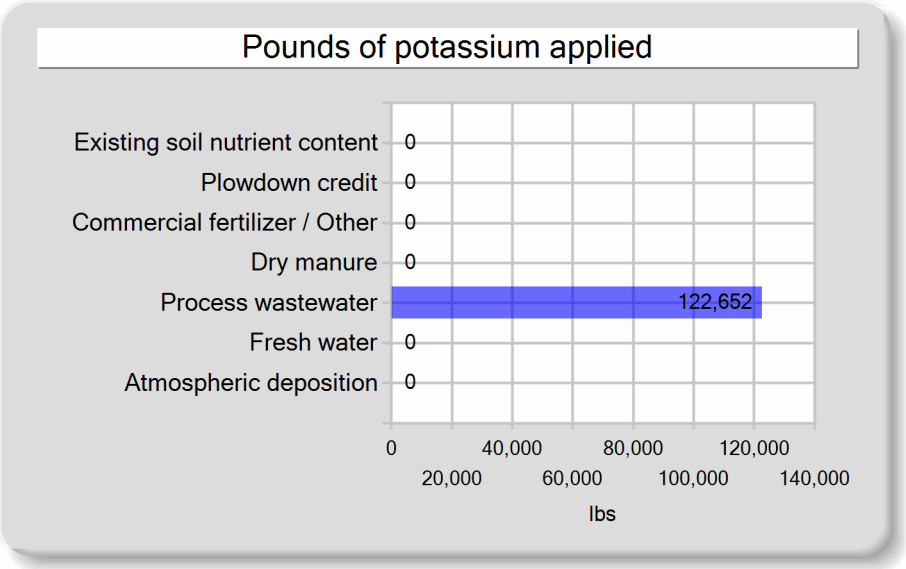
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

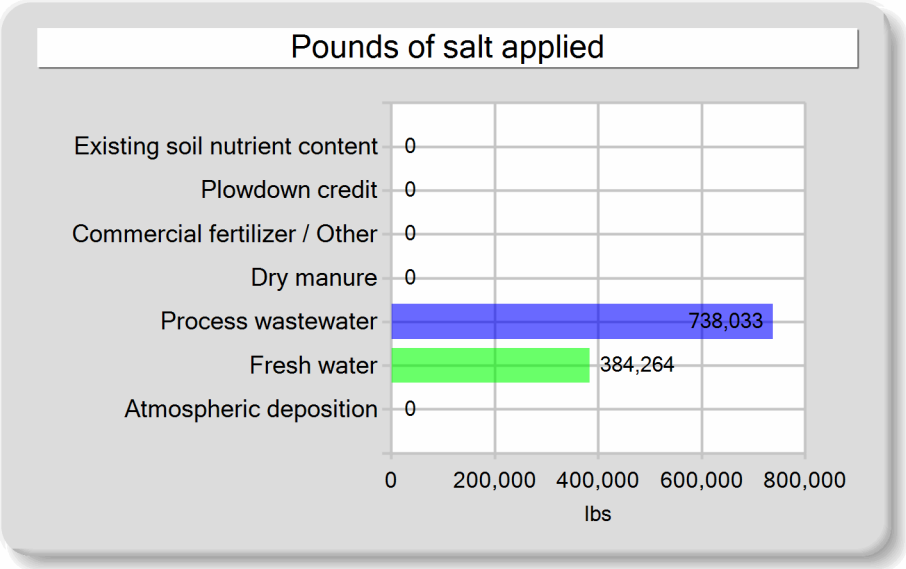


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Lloyd Fagundes Lloyd Fagundes (Jun 25, 2024 12:52 PDT)	Larry A. Matos Larry A. Matos (Jun 27, 2024 09:21 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Fagundes, Fagundes, Fagundes	California Dairy Farms-Bloss
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 25, 2024	Jun 27, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Matos Dairy
P.O. Box 1090
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: River Rock Ranches
RECEIVED DATE: 9/19/23 3:00 PM
SUBMITTED BY: KJ
ANALYZED DATE: 9/27/23

LAB SAMPLE ID: T3262044A
CUST SAMPLE ID: Newman
Plant Tissue F18 (F18)
SOURCE: Corn Silage

	As Received Basis	30% Dry Basis	100% Dry Basis
Moisture,% SM2540B	73.6	70.0	0.00
Dry Matter,% SM 2540B	26.4	30.0	100
Crude Protein,%	2.16	2.46	8.19
Total Nitrogen,% Combustion	0.346	0.393	1.31
Pounds of Nitrogen Removed per Ton	6.92	7.86	26.2
Ash,% ASTM	2.02	2.29	7.65
Total Phosphorus,% SM 4500P E	0.049	0.056	0.186
Total Potassium,% EPA 200.7	0.411	0.467	1.56

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Total Nitrogen (TN)				
Total Phosphorus (TP)				
Total Potassium (TK)				

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ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Matos Dairy
P.O. Box 1090
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: River Rock Ranches
RECEIVED DATE: 9/19/23 3:00 PM
SUBMITTED BY: KJ
ANALYZED DATE: 9/27/23

LAB SAMPLE ID: T3262044B
CUST SAMPLE ID: Redding E
Plant Tissue F18 (F18)
SOURCE: Corn Silage

	As Received Basis	30% Dry Basis	100% Dry Basis
Moisture,% SM2540B	67.6	70.0	0.00
Dry Matter,% SM 2540B	32.4	30.0	100
Crude Protein,%	3.20	2.96	9.88
Total Nitrogen,% Combustion	0.512	0.474	1.58
Pounds of Nitrogen Removed per Ton	10.2	9.48	31.6
Ash,% ASTM	2.61	2.42	8.06
Total Phosphorus,% SM 4500P E	0.068	0.063	0.211
Total Potassium,% EPA 200.7	0.609	0.564	1.88

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Total Nitrogen (TN)				
Total Phosphorus (TP)				
Total Potassium (TK)				

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Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: River Rock Ranches
RECEIVED DATE: 9/19/23 3:00 PM
SUBMITTED BY: KJ
ANALYZED DATE: 9/21/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 22.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW #9	T3262044C	Electrical Conductivity (EC)	EPA 120.1 and	1.36	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	52.2	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	780	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	55.0	mg/L
		Soluble Salts (SALT-SOL)	Calculation	868	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	96.6	94.0 (T3262039B)	98.8 (T3262039B)			4.97
Electrical Conductivity (EC)	99.3			0.464 (T3262039B)	0.462 (Dup)	0.475
Nitrate Nitrogen (NO3-N)	95.5	93.5 (T3262023A)	92.0 (T3262023A)			1.62
Total Dissolved Solids (TDS)	100			1500 (T3268021J)	1440 (Dup)	4.43
Total Nitrogen (TN)	97.4			2.80 (T3261032A)	2.80 (null)	0.000

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Compliance Analysis Report

Customer
Matos Dairy
P.O. Box 1090
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: River Rock Ranches
RECEIVED DATE: 9/19/23 3:00 PM
SUBMITTED BY: KJ
ANALYZED DATE: 9/21/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 21.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW #10	T3262044D	Electrical Conductivity (EC)	EPA 120.1 and	1.32	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	27.8	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	740	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	30.6	mg/L
		Soluble Salts (SALT-SOL)	Calculation	845	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	96.6	94.0 (T3262039B)	98.8 (T3262039B)			4.97
Electrical Conductivity (EC)	99.3			0.464 (T3262039B)	0.462 (Dup)	0.475
Nitrate Nitrogen (NO3-N)	95.5	93.5 (T3262023A)	92.0 (T3262023A)			1.62
Total Dissolved Solids (TDS)	100			1500 (T3268021J)	1440 (Dup)	4.43
Total Nitrogen (TN)	97.4			2.80 (T3261032A)	2.80 (null)	0.000

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Compliance Analysis Report

Customer
Matos Dairy
P.O. Box 1090
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: River Rock Ranches
RECEIVED DATE: 9/19/23 3:00 PM
SUBMITTED BY: KJ
ANALYZED DATE: 9/21/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Bloss DW	T3262044E	Electrical Conductivity (EC)	EPA 120.1 and	1.55	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.91	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	992	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	96.6	94.0 (T3262039B)	98.8 (T3262039B)	4.97
Electrical Conductivity (EC)	99.3			0.475
Nitrate Nitrogen (NO3-N)	95.5	93.5 (T3262023A)	92.0 (T3262023A)	1.62

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Compliance Analysis Report

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Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: River Rock Ranches
RECEIVED DATE: 9/19/23 3:00 PM
SUBMITTED BY: KJ
ANALYZED DATE: 9/21/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 22.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
FS Barn DW	T3262044F	Electrical Conductivity (EC)	EPA 120.1 and	1.19	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.60	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	764	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	96.6	94.0 (T3262039B)	98.8 (T3262039B)	4.97
Electrical Conductivity (EC)	99.3			0.475
Nitrate Nitrogen (NO3-N)	97.0	99.0 (W3264003A)	99.6 (W3264003A)	0.604

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Customer
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Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: River Rock Ranches
RECEIVED DATE: 9/19/23 3:00 PM
SUBMITTED BY: KJ
ANALYZED DATE: 9/21/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 22.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Hosp. Barn	T3262044G	Electrical Conductivity (EC)	EPA 120.1 and	1.12	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	3.54	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	714	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	96.6	94.0 (T3262039B)	98.8 (T3262039B)	4.97
Electrical Conductivity (EC)	99.3			0.475
Nitrate Nitrogen (NO3-N)	97.0	99.0 (W3264003A)	99.6 (W3264003A)	0.604

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Matos Dairy
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Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: River Rock Ranches
RECEIVED DATE: 9/19/23 3:00 PM
SUBMITTED BY: KJ
ANALYZED DATE: 9/21/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Milk Barn	T3262044H	Electrical Conductivity (EC)	EPA 120.1 and	1.32	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	3.81	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	843	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	96.6	94.0 (T3262039B)	98.8 (T3262039B)	4.97
Electrical Conductivity (EC)	99.3			0.475
Nitrate Nitrogen (NO3-N)	97.0	99.0 (W3264003A)	99.6 (W3264003A)	0.604

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: River Rock Ranches
Mailing address: Amatos Dairy
(not facility address)
Phone#:

Authorized Copy Released To: F & R Ag Services

Sampled By: K.Jones
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	9/14	8:30	Corn silage	Newman						✓	
2	9/13	4:45	↓	Redding ↓						✓	
3	9/18	8:00	Irrigation well	IW#9		22.6					H7
4	9/18	8:15	↓	IW#10		21.8					H7
5	9/18	8:30	Domestic well	Bloss DW	✓	21.6					
6	|	8:45	↓	FS Barn DW	✓	22.0					
7	|	9:00	↓	Hosp. Barn	✓	22.0					
8	|	9:15	↓	Milk Barn	✓	21.8					
9	↓										
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st		F & R Ag Services				
2nd		DENELE ANALYTICAL, INC.	9/23 3:00			T3262044

Required to be submitted to any agency? (YES / NO)

2023 Annual Report- River Rock Ranches

Final Audit Report 2024-06-27

Created:	2024-06-19
By:	Kristen Jones (kjones@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAv03enjfiHl6XY4e8RPdFhVUcj2QRFu6r

"2023 Annual Report- River Rock Ranches" History

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Signer amacias@fagundesdairy.com entered name at signing as Lloyd Fagundes
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Signer jmatos@matosfarms.com entered name at signing as Larry A. Matos
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